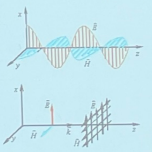
- Inserting (2.14) into (2.10) gives

$$\nabla\cdot\mathbf{E} = i\mathbf{k}\cdot\mathbf{E}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)] = 0 \qquad (2.17)$$

- This means that $\mathbf{E}_0 \perp \mathbf{k}$ $\qquad \mathbf{H}_0 = (\omega\mu_0)^{-1}\mathbf{k}\times\mathbf{E}_0$ (2.18)
- From (2.7) we have for plane waves: $\mathbf{H} = (\omega\mu_0)^{-1}\mathbf{k}\times\mathbf{E} = \mathbf{H}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)]$ (2.19)
- Furthermore we have with (2.1e,f) and (2.4):

$$\mathbf{D} = \mathbf{D}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)] = \varepsilon_0\mathbf{E}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)]$$

$$\mathbf{B} = \mathbf{B}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)] = \omega^{-1}\mathbf{k}\times\mathbf{E}_0\exp[i(\mathbf{k}\cdot\mathbf{r}-\omega t)]$$



- So we have : $\mathbf{D}\perp\mathbf{k}$, $\quad\mathbf{k}\perp\mathbf{B}$, $\quad\mathbf{B}\perp\mathbf{D}$
- In vacuum and isotropic media one has:

$\mathbf{E}//\mathbf{D}$ and $\mathbf{H}//\mathbf{B}$

- The momentum density of the electromagnetic field is given by

$\mathbf{\Pi} = \mathbf{D}\times\mathbf{B} \qquad \mathbf{\Pi}//\mathbf{k}$

- The energy flux density, Poynting vector : $\mathbf{S} = \mathbf{E}\times\mathbf{H}$
- In vacuum and isotropic materials: $\mathbf{S}//\mathbf{\Pi}$
- The average value of Poynting vector is usually called the intensity I
- For the plane monochromatic waves, we obtain:

$$\langle\mathbf{S}\rangle = \frac{1}{2}|\mathbf{E}\times\mathbf{H}| = \frac{1}{2}\frac{1}{c\mu_0}\mathbf{E}_0^2 = \frac{1}{2}\frac{c}{\mu_0}\mathbf{B}_0^2 = \frac{1}{2}c\mu_0\mathbf{H}_0^2$$

2.2 Electromagnetic Radiation in Vacuum

- In vacuum the following conditions are fulfilled (2.6)

$\boldsymbol{P} = 0;\ \boldsymbol{M} = 0;\ \boldsymbol{\rho} = 0;\ \boldsymbol{j} = 0.$

- With the help of (2.1e,f) this simplifies (2.1c,d) to (2.7a,b)

$\nabla\times\boldsymbol{E} = -\mu_0\dot{\boldsymbol{H}}$ and $\nabla\times\boldsymbol{H} = \varepsilon_0\dot{\boldsymbol{E}}$

- Applying $\nabla\times$ to (2.7a) and $\partial/\partial t$ to (2.7b) yields (2.8)

$\nabla\times(\nabla\times\mathbf{E}) = -\mu_0\nabla\times\dot{\mathbf{H}} \qquad \nabla\times\dot{H} = \varepsilon_0\ddot{E}$

- With the properties: $\nabla\times(\nabla\times\mathbf{F}) = \nabla(\nabla\cdot\mathbf{F}) - \nabla^2\mathbf{F}$, we find: (2.9)

$$-\mu_0\varepsilon_0\ddot{\mathbf{E}} = \nabla\times(\nabla\times\mathbf{E}) = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E} \qquad (2.10)$$

$$\nabla\cdot\mathbf{E} = 0 \qquad (2.11)$$

- So we get: $\nabla^2\mathbf{E} - \mu_0\varepsilon_0\ddot{\mathbf{E}} = 0$
- For the most simple ones, namely on plane waves $\mathbf{E}(\mathbf{r},\mathbf{t}) = \mathbf{E}_0\exp[i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)]$ (2.14)
- $\mathbf{E}_0$ is the amplitude, $\boldsymbol{k}$ is the wave vector and ω is the angular frequency.

$$k = |\boldsymbol{k}| = \frac{2\pi}{\lambda_v} \qquad \frac{\omega}{k} = \left(\frac{1}{\mu_0\varepsilon_0}\right)^{\frac{1}{2}} = c$$

Helmholtz Equation $(\nabla^2 + k^2)E(\bar{r}) = 0$

◆ Actually, all matter has some diamagnetism. And this is a rather small effect of the order of 10^{-6}, which have negligible influence on the optical properties.

◆The current is driven by the electric field: $\mathbf{j} = \sigma\mathbf{E}$, and σ is small for intrinsic or weakly doped semiconductors. So $|\mathbf{j}| = |\sigma\mathbf{E}| << |\dot{\mathbf{D}}|$.

◆We neglect $\boldsymbol{j}$ for weakly doped semiconductors , and get :

$$\nabla^2\mathbf{E} - \mu_0\varepsilon_0\ddot{\mathbf{E}} = \mu_0\ddot{\mathbf{P}} \qquad (2.26)$$

Equation (2.26) states the well-known fact that every dipole p and polarization P with a non vanishing second derivative in time radiates an electromagnetic wave.

? The relationships between ***D***, ***E*** and ***P***

$$\boxed{\begin{aligned}\mathbf{P} &= \varepsilon_0\chi\mathbf{E}\\ \mathbf{D} &= \varepsilon_0(1+\chi)\mathbf{E} = \varepsilon\varepsilon_0\mathbf{E}\\ \varepsilon &= (1+\chi)\end{aligned}} \qquad (2.27)$$

◆ Now we make a very important assumption.

The quantities ε and χ are called the dielectric function and the susceptibility, respectively.

1. Both of them are tensors, But for simplicity, can be treated as scalar quantities.
2. When frequency is dominant, we can drop k
3. $\varepsilon(0)$ is usually called the dielectric constant

$\varepsilon = \varepsilon(\omega,\mathbf{k})$

$\chi = \chi(\omega,\mathbf{k}) = \varepsilon(\omega,\mathbf{k}) - 1$

$\varepsilon(\omega,\mathbf{k}) = \varepsilon_1(\omega,\mathbf{k}) + i\varepsilon_2(\omega,\mathbf{k})$

30

$$\nabla^2\mathbf{E} - \mu_0\varepsilon_0\ddot{\mathbf{E}} = \mu_0\ddot{\mathbf{P}} \longrightarrow \nabla^2\mathbf{E} - \mu_0(\omega)\varepsilon_0\varepsilon(\omega)\ddot{\mathbf{E}} = 0$$

If magnetic properties are included $\nabla^2\mathbf{E} - \mu_0\mu(\omega)\varepsilon_0\varepsilon(\omega)\ddot{\mathbf{E}} = 0$ Mostly, $\mu(\omega) \approx 1$

For simplicity, in the case of plane harmonic waves: $\mathbf{E}(\mathbf{r},\mathbf{t}) = \mathbf{E}_0\exp[\mathrm{i}(\mathbf{k}\cdot\mathbf{r}-\omega t)]$ (2.32)

Inserting into the above, and reads:

$$\frac{c^2k^2}{\omega^2} = \varepsilon(\omega) \qquad k = \frac{\omega}{c}\varepsilon^{\frac{1}{2}}(\omega) = \frac{2\pi}{\lambda_v}\varepsilon^{\frac{1}{2}}(\omega) = k_v\varepsilon^{\frac{1}{2}}(\omega)$$

For simplicity, we introduce the complex index of refraction: $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega) = \varepsilon^{\frac{1}{2}}(\omega)$

So in matter, the light with k can be complex quantity :

$$k = \frac{\omega}{c}\tilde{n}(\omega) = \boxed{\frac{\omega}{c}n(\omega)} + i\frac{\omega}{c}\kappa(\omega) = \frac{2\pi}{\lambda_v}\tilde{n}(\omega) = k_v\tilde{n} \qquad (2.36)$$

Describes the oscillatory part of the wave .

Insertion (2.36) into (2.32): $\mathbf{E}(\mathbf{r},\mathbf{t}) = \mathbf{E}_0\exp\left\{\mathrm{i}\left[\frac{\omega}{c}n(\omega)\hat{\mathbf{k}}\cdot\mathbf{r}-\omega t\right]\right\}\exp\left[-\frac{\omega}{c}\kappa(\omega)\hat{\mathbf{k}}\cdot\mathbf{r}\right]$ (2.37)

$\hat{\mathbf{k}}$ is the unit vector of $\mathbf{k}$

The oscillatory spatial propagation of light in matter

A damping of light in the direction of propagation, called absorption

Ch3

- 3.2 Microscopic Aspects
- **3.2.1 Absorption, stimulated and spontaneous emission, virtual excitation** 要懂

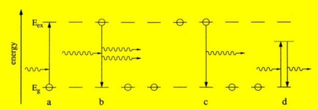


Fig. 3.16. Schematic representations of various interaction processes of light with matter namely absorption (a), stimulated emission (b), spontaneous emission (c) and virtual excitation (d)

- Two-level "atoms"

Fig.3.16a: Absorption process $\hbar\omega = E_{ex} - E_g$

Fig.3.16b: Stimulated emission,

Basic mechanism for all lasers (light amplification by stimulated emission of radiation).

Fig.3.16c: Spontaneous emission, an electron in the excited state transit to the ground state by itself.

Fig.3.16d: Virtual excitation: with an energy different from the eigen-energy of this excited state.

Uncertainty principle of quantum mechanics :

$$\Delta x_i\Delta p_i \geq h \quad for \quad i = 1,2,3. \qquad (3.37a)$$

$$\Delta E\Delta t \approx \hbar \qquad (3.37b)$$

From 3.37b , it is possible to violate energy conservation by an amount ΔE up until a maximum time Δt .

- To conclude this section: the net rate in Fig.3.16a-c : 可能考计算，速率方程

$$\frac{\partial N_{ph}}{\partial t} = \boxed{-N_A\alpha_g N_{ph}\left|H^D_{g\to e}\right|^2} + \boxed{N_A(1-\alpha_g)(1+N_{ph})\left|H^D_{e\to g}\right|^2} \qquad (3.61)$$

Absorption $\qquad$ Spontaneous and stimulated emission

In which α_g is a fraction in the ground state, $(1-\alpha_g)$ are in the excited state.

From (3.55) : $\left|H^D_{g\to e}\right|^2 = \left|H^D_{e\to g}\right|^2 = \left|H^D\right|^2$ (3.62)

and hence:

$$\frac{1}{|H^D|^2}\frac{\partial N_{ph}}{\partial t} = \boxed{N_{ph}N_A(1-2\alpha_g)} + \boxed{N_A(1-\alpha_g)} \qquad (3.63)$$

Linearly depends on N_{ph}
Absorption and stimulated emission

Independent of N_{ph}
Spontaneous emission

1. For net absorption : $\alpha_g > \frac{1}{2}$
2. Amplification or optical gain : $\alpha_g < \frac{1}{2}$
3. The semiconductor becomes transparent $\alpha_g = \frac{1}{2}$
4. The situation $\alpha_g < \frac{1}{2}$ can't be reached in thermal equilibrium, but only under a pump source

CH4

Lorentz Oscillator Model

非常重要

- The electric field of incident light on the independent oscillators:

$$\boldsymbol{E} = (E_0, 0, 0)\exp\left[i(k_z z - \omega t)\right] \quad (4.4a)$$

Considering the oscillator at $z = 0$:

$$\boldsymbol{E} = (E_0, 0, 0)e^{-i\omega t} \quad (4.4b)$$

Assume every oscillator carries a charge e, then a force by $\boldsymbol{E}$:

$$F = eE_0 e^{-i\omega t} \quad (4.5)$$

The equation of motion is:

$$m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \quad (4.6)$$

The general solution of (4.6) =the general solution of the corresponding homogeneous equation + a special solution of the inhomogeneous one.

Assume : $x(t) = \boxed{x_0 \exp\left[-i\left(\omega_0'^2 - \gamma^2/4\right)^{1/2} t\right]\exp(-t\gamma/2)} + \boxed{x_p e^{-i\omega t}}$ (4.7)

For $\omega_0'^2 - \gamma^2/4 > 0$: defines the regime of weak damping.

The solution of the homogeneous equation describes a transient feature 齐次,瞬态解

The second part in Eq (4.7) is a forced oscillation, substitute it into (4.6): 稳态解

$$x_p = \frac{eE_0}{m}\left(\omega_0'^2 - \omega^2 - i\omega\gamma\right)^{-1} \quad (4.9)$$

振幅

The dipole moment :

$$p_x = ex_p \quad (4.10)$$

偏极化

P5 — P14 都重要

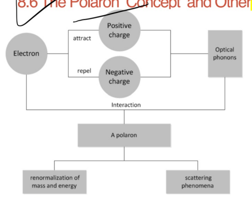
CH7

光学,声学声子、布里渊区

CH8. 有效质量, 8.6 (P15)

概念

8.6 The Polaron Concept and Other Electron–Phonon Interaction Processes



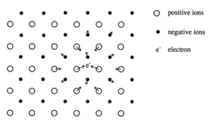


Fig. 8.8. The lattice distortion around a carrier in a (partly) ionic semiconductor illustrating the polaron concept

- The effective mass $m_{e,h}$ of a polaron is greater than that of an electron in a rigid lattice $m_{e,h}^{r.l.}$:

$$m_{e,h} \simeq m_{e,h}^{r.l.}\left(1 + \frac{\alpha_{e,h}}{6}\right) \quad (8.19a)$$

$$\alpha_{e,h} = \frac{e^2}{8\pi\varepsilon_0\hbar\omega_{LO}}\left(\frac{2m_{e,h}^{r.l.}\omega_{LO}}{\hbar}\right)\left(\frac{1}{\varepsilon_b} - \frac{1}{\varepsilon_s}\right) \quad (8.19b)$$

Additionally the lattice relaxation leads to a decrease of the width of the gap by :

$$\Delta E_g^{e,h} = \alpha_{e,h}\hbar\omega_{LO} \quad (8.19d)$$

The radius of the phonon cloud in the polaron :

$$r_p^{e,h} = \left(\frac{\hbar}{2m_{e,h}^{r.l.}\omega_{LO}}\right)^{\frac{1}{2}} \quad (8.19e)$$

- An electron or hole can be scattered inelastically:

(8.20a)

8.10 一定要懂

激子 概念

CH9 P9—P11

三维 与 二维 激子

CH10

The equation of motion for free carriers for an electric field :

$$m\ddot{x} = F_x = -\frac{m}{\tau}\dot{x} + eE_x^0 e^{-i\omega t} \qquad m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \tag{10.4a}$$

Resulting in a frequency-dependent electrical conductivity:

$$\sigma(\omega) = \sigma_0 \frac{1}{1 - i\omega\tau} \tag{10.4b}$$

and to a total dielectric function:

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \chi_{free\ \ carrier}(\omega) \tag{10.4c}$$

with the help of Maxwells equation $\nabla \times \mathbf{H} = \boldsymbol{j} + (\varepsilon_0 \dot{\mathbf{E}} + \dot{\mathbf{P}})$ $\quad \mathbf{P} = \varepsilon_0 \chi \mathbf{E}$ $\quad \nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}} = \boldsymbol{\sigma}\mathbf{E} + \varepsilon_0 \varepsilon_{Loretz} \dot{\mathbf{E}}$

we obtain :

$$\chi_{free\ \ carrier}(\omega) = \frac{i}{\varepsilon_0 \omega} \sigma(\omega)$$

and for the total dielectric function:

$$\varepsilon(\omega) = \varepsilon_b + \frac{\frac{ne^2}{m\varepsilon_0}}{-\omega^2 - i\omega\gamma}$$

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{i}{\varepsilon_0 \omega}\sigma(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{\omega_{PL}^{0^2}}{-\omega^2 - i\omega/\tau}$$

Quantized (surface) plasmon modes exist in small metal spheres or colloids.

CH11

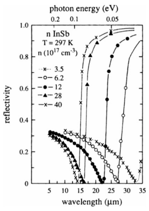
基本的散射

瑞利 布里渊、拉曼

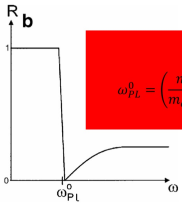
CH12

一定要会

- Plasmons as collective excitations of the carriers in a partly filled band usually do not exist in intrinsic semiconductors under thermodynamic equilibrium
- Two conditions under which plasmons can be observed, high doping or high excitation
  - In highly doped semiconductors, there is a high density either of electrons or of holes
  - In highly excited semiconductors, an electron–hole plasma can be formed, which consists of electrons and holes



The reflection spectrum in the vicinity of the plasmon resonance in doped InSb samples



$$\omega_{PL}^{0} = \left(\frac{ne^2}{m_e \varepsilon \varepsilon_0}\right)^{\frac{1}{2}}$$

CH13 直接、间接带隙的吸收

Absorption coefficient

$$\alpha(\hbar\omega) = A\sum W_{\mathrm{if}}^{\mathrm{ab}} n_i n_f' \quad (13.2.8)$$

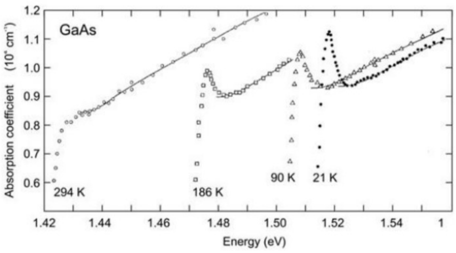
$$\alpha_d(\hbar\omega) = \begin{cases} A(\hbar\omega - E_g)^{1/2}, & \hbar\omega \geqslant E_g \\ 0, & \hbar\omega < E_g \end{cases} \quad (13.2.9a)$$

$$A \approx \frac{(2m_r^*)^{3/2} e^2}{\eta c m_e^* \hbar^2 \varepsilon_0} \quad (13.2.9b)$$

$$m_r^* = \frac{m_h^* m_e^*}{m_h^* + m_e^*} \quad (13.2.9c)$$

$m_e^* = m_e^* = m_0$, $\eta = 4$, $A \approx 2\times10^4\ \mathrm{cm}^{-1}$

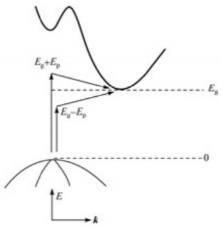
$$\alpha_d(\hbar\omega) = \begin{cases} \approx 2\times10^4(\hbar\omega - E_g)^{1/2}, & \hbar\omega \geqslant E_g \\ = 0, & \hbar\omega < E_g \end{cases} \quad (13.2.10)$$



Absorption spectra of GaAs close to the bandgap at different temperatures

(b) Indirect transition

- The band extrema do not occur at the same wave vector k, the material is referred to as an indirect-gap semiconductor.
- For momentum conservation in such semiconductors, optical transitions (absorption and photoluminescence) are indirect, a participation of an extra particle, i.e., phonon, is required.
- The probability of optical transitions (absorption and luminescence) is substantially lower, compared with direct transitions. Therefore, fundamental absorption in indirect-gap semiconductors is usually weaker as compared with direct-gap materials.



Absorption with phonon emission

$$\hbar\omega_e = E_c(\boldsymbol{k}_c) - E_v(\boldsymbol{k}_v) + E_p \quad (13.2.11a)$$

$$\hbar\omega \leqslant E_g + E_p \text{ 时 } \alpha_e(\hbar\omega) = 0 \quad (13.2.11b)$$

Absorption with phonon absorption

$$\hbar\omega_a = E_c(\boldsymbol{k}_c) - E_v(\boldsymbol{k}_v) - E_p \quad (13.2.12a)$$

$$\hbar\omega \leqslant E_g - E_p \text{ 时 } \alpha_a(\hbar\omega) = 0 \quad (13.2.12b)$$

Absorption coefficient with phonon emission

$$\alpha_e(\hbar\omega) = \begin{cases} \dfrac{B(\hbar\omega - E_g - E_p)^2}{1-\exp(-E_p/k_BT)}, & \hbar\omega > (E_g + E_p) \\ 0, & \hbar\omega \leqslant (E_g + E_p) \end{cases} \quad (13.2.13)$$

Absorption coefficient with phonon absorption

$$\alpha_a(\hbar\omega) = \begin{cases} \dfrac{B(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_BT)-1}, & \hbar\omega > (E_g - E_p) \\ 0, & \hbar\omega \leqslant (E_g - E_p) \end{cases} \quad (13.2.14)$$

The total absorption coefficient with phonon emission and absorption

$$\alpha_i(\hbar\omega) = \begin{cases} B\left[\dfrac{(\hbar\omega - E_g - E_p)^2}{1-\exp(-E_p/k_BT)} + \dfrac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_BT)-1}\right], & \hbar\omega > (E_g + E_p) \\ B\dfrac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_BT)-1}, & E_g - E_p \leqslant \hbar\omega \leqslant E_g + E_p \\ 0, & \hbar\omega \leqslant (E_g - E_p) \end{cases} \quad (13.2.15)$$

和激子相关的吸收

和激子相关的吸收

➢The absorption spectra:

- 13.9a an absorption spectrum of ZnO layer at RT
- 13.9b experimental transmission spectra are shown A and $B\Gamma_5$ resonance of a thin CdS platelet type sample
- 13.9c shows a calculated transmission spectrum of the $A\Gamma_5$ $n_B = 1$ resonance
- 13.9d overview of the absorption spectrum of a thin GaAs sample.

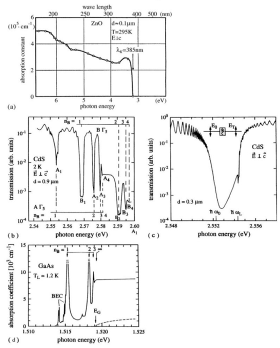
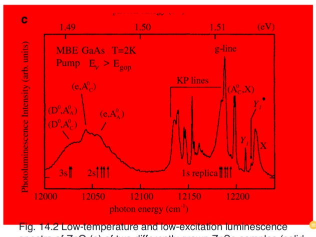


Fig. 13.9. An experimental absorbtion spectrum of ZnO at RT (**a**), an experimental transmission spectrum of CdS for the orientation $\boldsymbol{E} \perp \boldsymbol{c}$ in the region of the A and B exciton resonances (**b**); a calculated one for the $A\Gamma_5$ resonance for $n_B = 1$ (**c**); and an absorption spectrum for a thin GaAs sample (**d**). According to [43M1] to [79V1, 82R1] and to [91U1], respectively

➢ BEC often show up in luminescence and absorption spectra as extremely narrow peaks.

➢ Chemical shift or central cell correction：a splitting of the lines caused by the chemical nature of the binding atom.

➢ the low temperature luminescence of high quality samples is generally dominated by defect (or localiation-) related recombination processes.



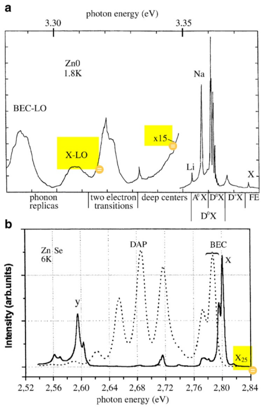


Fig. 14.2 Low-temperature and low-excitation luminescence spectra of ZnO (a) of two differently grown ZnSe samples (solid line: MBE growth with element sources, dashed line: with a compound source) (b) and of GaAs (c)

CH14 只有14.3不考

CH15 除了15.4都考

CH16 STARK EFFECT

CH17 好像没有